Task 1 — Rakuten Product Classification: EDA & Preprocessing

1. Project Introduction

Context

This notebook addresses the **Rakuten France Multimodal Product Data Classification** challenge (ENS ChallengeData #35). The task is to automatically assign one of 27 product category codes (`prdtypecode`) to e-commerce listings using three information sources:

- `designation` : short product title (always present)
- `description` : longer HTML-rich text (missing in ~35% of rows)
- **Product image**: JPEG photograph (always present)

Economic context: Manual product cataloging is a bottleneck at scale. Accurate automated classification enables better search, recommendation, and inventory management.

Scientific context: This is a multimodal classification problem requiring fusion of NLP (text) and computer vision (image) signals — an active research area.

Objectives

1. Understand the dataset (volume, quality, class distribution, missing data).
2. Produce ≥5 annotated visualizations with statistical validation.
3. Preprocess text and images so the dataset is ready for Rendu 2 modeling.
4. Save cleaned training and test sets as pickle files.

Target

Predict `prdtypecode` (27 categories) from `designation` + `description` + product image. Primary evaluation metric: **weighted F1-score (macro)**.

2. Data Understanding

```
Training set : 84,916 rows × 6 columns
Test set     : 13,812 rows × 4 columns

Training columns : ['designation', 'description', 'productid', 'imageid', 'prdtypecod
e', 'category']

Target classes   : [10, 40, 50, 60, 1140, 1160, 1180, 1280, 1281, 1300, 1301, 1302, 1
320, 1560, 1920, 1940, 2060, 2220, 2280, 2403, 2462, 2522, 2582, 2583, 2585, 2705, 29
05]
Number of classes: 27

Category mapping:
  10 — Livres
  40 — Jeux vidéo PC/consoles
  50 — Accessoires gaming
  60 — Consoles de jeux vidéo
  1140 — Produits dérivés/figurines geek
  1160 — Cartes collectionnables
  1180 — Figurines jeux de société
  1280 — Jouets/peluches/poupées
  1281 — Jeux de société/cartes
  1300 — Petites voitures/maquettes
  1301 — Accessoires petite enfance/jeux
  1302 — Jeux d'extérieur
  1320 — Sacs femmes/accessoires enfance
  1560 — Mobilier/déco maison
  1920 — Linge de maison
  1940 — Nourriture/épicerie
  2060 — Lampes/accessoires déco
  2220 — Accessoires animaux
  2280 — Magazines
  2403 — Livres et BDs
  2462 — Consoles/jeux vidéo
  2522 — Papeterie/rangement bureau
  2582 — Mobilier extérieur/accessoires
  2583 — Accessoires piscine
  2585 — Outillage/jardinage
  2705 — BDs et livres
  2905 — Jeux en téléchargement

(84916, 4) (84916, 1) (13812, 4)
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 84916 entries, 0 to 84915
Data columns (total 6 columns):
 #   Column       Non-Null Count  Dtype 
---  ------       --------------  ----- 
 0   designation  84916 non-null  object
 1   description  55116 non-null  object
 2   productid    84916 non-null  int64 
 3   imageid      84916 non-null  int64 
 4   prdtypecode  84916 non-null  int64 
 5   category     84916 non-null  object
dtypes: int64(3), object(3)
memory usage: 4.5+ MB

--- Missing values ---
designation        0
description    29800
productid          0
imageid            0
prdtypecode        0
category           0
dtype: int64

Description missing rate: 35.1%

Sample rows:
```

	designation	description	productid	imageid	prdtypecode	category
0	Olivia: Personalisiertes Notizbuch / 150 Seite...	NaN	3804725264	1263597046	10	10 — Livres
1	Journal Des Arts (Le) N° 133 Du 28/09/2001 - L...	NaN	436067568	1008141237	2280	2280 — Magazines
2	Grand Stylet Ergonomique Bleu Gamepad Nintendo...	PILOT STYLE Touch Pen de marque Speedlink est ...	201115110	938777978	50	50 — Accessoires gaming

Key observations

- **84,916 training samples**, 27 product categories, 13,812 unlabeled test samples.
- `designation` is always present; `description` is missing in ~35% of rows → image and designation are the only reliable signals for a large fraction of the catalog.
- `x_train.csv` (features) merged with `y_train.csv` (labels) on the shared index to build the labeled training set.
- Both `productid` and `imageid` are identifiers used to locate images — not model features.

3. Visualizations & Statistics

```
Chi-squared test vs uniform distribution:
  χ² = 36570.3,  p-value = 0.00e+00
  → Significant imbalance (α=0.05)

Most frequent class : 2583 — Accessoires piscine (10,209 samples, 12.0%)
Least frequent class: 1180 — Figurines jeux de société (764 samples, 0.9%)
```

Fig 1 — Business insight: The catalog is heavily imbalanced. A few categories dominate while niche product types have far fewer examples. This requires class-weighted loss functions or oversampling strategies in Rendu 2. The chi-squared test confirms the imbalance is statistically significant ($p \ll 0.05$).

Recommendation:

- Apply **class-weighted training** (`class_weight='balanced'` in sklearn, `class_weight` in PyTorch loss) to compensate for imbalance.
- Consider **stratified train/validation splits** to ensure all 27 classes are represented in evaluation.
- For the smallest categories, evaluate per-class F1 separately — weighted-F1 can mask poor performance on rare classes.

```
Chi-squared test of independence (category vs description missingness):
  χ² = 43757.8,  dof = 26,  p-value = 0.00e+00
  → Non-random missingness (α=0.05)
```

Fig 2 — Business insight: Missing description rates vary dramatically across categories (from near 0% to >80%). Categories with high missing rates are effectively image-only, making visual features critical for those classes. The chi-squared test of independence confirms missingness is strongly associated with product type ($p \ll 0.05$) — not random.

Recommendation:

- For image-heavy categories (missing rate > 80%), prioritize **image-based features** in the multimodal pipeline — text alone will be insufficient.
- Fill missing descriptions with an empty string (not `designation`) to preserve the distinction between "no description" and "short description".
- Consider adding a **`has_description` binary feature** — the missingness pattern itself is a strong categorical signal.

```
Pearson  r = -0.015  (p = 0.9416)
Spearman ρ = 0.038  (p = 0.8513)
```

Fig 2b — Business insight:
This scatter plot tests the hypothesis that categories with higher missing description rates attract fewer products.

Results: Pearson r = −0.015 (p = 0.94), Spearman ρ = 0.038 (p = 0.85) — the hypothesis is **not supported**. There is essentially no correlation between missing description rate and category size.

The distribution is bimodal on the x-axis: categories either have near-zero missing rates (e.g. books, always well-described) or very high missing rates (80–100%), suggesting this is a structural merchant convention per category, not a signal of low adoption. Category size is driven by product type popularity, not description completeness.

Recommendation:

- Do **not** use missing description rate as a proxy for category size or data quality.
- Instead, treat each category's missingness pattern independently — some categories are structurally image-only and should be handled by the image branch of the multimodal model.

```
Kruskal–Wallis test (combined word count across categories):
  H = 34696.6,  p–value = 0.00e+00
```

Fig 3 — Business commentary: Text length varies significantly across categories. Some product types have long descriptions while others rely on very short titles. The Kruskal-Wallis test confirms word count distributions differ significantly across categories ($p \ll 0.05$), meaning text length is a useful feature.

Recommendation:

- Include **word count and character count** as explicit numeric features alongside text embeddings — they carry discriminative signal independent of vocabulary.
- For text-based models (TF-IDF, BERT), avoid truncating very long descriptions too aggressively; categories with rich descriptions benefit from full context.
- Short-text categories may benefit more from image features than from text embeddings.

Fig 4 — Business commentary: Visual inspection confirms categories are often visually distinguishable (e.g. toys vs books vs clothing), but within-category variation exists (different backgrounds, scales, angles). Image features alone will not suffice for all classes — text signals are essential for disambiguation. No formal statistical test applies to visual inspection.

Recommendation:

- Use **data augmentation** (random crop, flip, color jitter) during CNN training to improve robustness to the within-category visual variation observed here.
- Consider a **late fusion** approach: train a text classifier and an image classifier independently, then combine their softmax outputs — this allows each modality to specialise.
- For categories where images are ambiguous (e.g. generic white-background product shots), rely on text features as a tiebreaker.

```
Top 5 TF-IDF terms per category (first 5):
  10 — Livres: [('tome', 0.0319), ('edition', 0.0251), ('guide', 0.0181), ('histoir
e', 0.0164), ('book', 0.0123)]
  40 — Jeux vidéo PC/consoles: [('import', 0.1002), ('edition', 0.0421), ('japonais',
0.0413), ('version', 0.0324), ('jeu', 0.0322)]
  50 — Accessoires gaming: [('nintendo', 0.0767), ('ps4', 0.0587), ('manette', 0.053
2), ('console', 0.0478), ('jeu', 0.0475)]
  60 — Consoles de jeux vidéo: [('console', 0.1732), ('oui', 0.1616), ('jeu', 0.141
1), ('console jeu', 0.111), ('nom marque', 0.1037)]
  1140 — Produits dérivés/figurines geek: [('figurine', 0.1695), ('pop', 0.0513), ('s
tar', 0.036), ('shirt', 0.0339), ('wars', 0.0339)]
```

Fig 5 — Business commentary: TF-IDF reveals strongly discriminative vocabulary per category — words common within a class but rare globally (e.g. category-specific brand names, product types, French nouns). High-scoring TF-IDF terms are statistically validated discriminators, confirming text features will be highly informative for classification.

Recommendation:

- Use the top TF-IDF terms per category as a **sanity check for label quality** — if the top terms for a category are unrelated to the expected product type, it may indicate label noise.
- During feature engineering, consider building **category-specific vocabulary masks** to boost the weight of discriminative terms in the TF-IDF matrix.
- For deep learning models, **French-specific pretrained embeddings** (CamemBERT, FlauBERT) will outperform generic multilingual models given the predominantly French vocabulary.

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
Cell In[12], line 15
     13 widths = np.array(widths)
     14 heights = np.array(heights)
---> 15 print(f"Width  — mean: {widths.mean():.0f} px, std: {widths.std():.0f}, min:
{widths.min()}, max: {widths.max()}")
     16 print(f"Height — mean: {heights.mean():.0f} px, std: {heights.std():.0f}, mi
n: {heights.min()}, max: {heights.max()}")
     18 from plotly.subplots import make_subplots

File ~/.pyenv/versions/3.12.0/lib/python3.12/site-packages/numpy/_core/_methods.py:4
7, in _amin(a, axis, out, keepdims, initial, where)
     45 def _amin(a, axis=None, out=None, keepdims=False,
     46           initial=_NoValue, where=True):
---> 47     return umr_minimum(a, axis, None, out, keepdims, initial, where)

ValueError: zero-size array to reduction operation minimum which has no identity
```

Fig 6 — Business commentary: Product images are not uniform in size. For CNN-based models in Rendu 2, all images must be resized to a fixed resolution (224×224 px, standard for pretrained models like ResNet/EfficientNet). Aspect ratio distortion is acceptable since product images are typically centered and square-framed.

Recommendation:

- Resize all images to **224×224 px** before training (standard input for ResNet, EfficientNet, ViT pretrained on ImageNet).
- Apply **center-crop** rather than stretch resize to preserve aspect ratios where possible.
- Normalise pixel values using ImageNet mean/std (`mean=[0.485, 0.456, 0.406]` , `std=[0.229, 0.224, 0.225]`) when using pretrained CNNs.

4. Preprocessing & Feature Engineering

```
Text preprocessing complete.
Sample cleaned text (row 0): olivia personalisiertes notizbuch seiten punktraster ca
din a rosen design

has_description rate (train): 64.9%

Missing train images: 84916 / 84916
```

```
---------------------------------------------------------------------------
FileNotFoundError                         Traceback (most recent call last)
Cell In[14], line 7
      4 print(f"Missing train images: {len(missing_train)} / {len(train)}")
      6 sample_row = train.iloc[0]
----> 7 img_original = Image.open(get_image_path(sample_row))
      8 img_resized = img_original.resize((224, 224))
     10 fig_resize = make_subplots(
     11     rows=1, cols=2,
     12     subplot_titles=[
   (...)     15     ],
     16 )

File ~/.pyenv/versions/3.12.0/lib/python3.12/site-packages/PIL/Image.py:3493, in open
(fp, mode, formats)
   3491 if is_path(fp):
   3492     filename = os.fspath(fp)
-> 3493     fp = builtins.open(filename,     )
   3494     exclusive_fp = True
   3495 else:

FileNotFoundError: [Errno 2] No such file or directory: '../images/image_train/image_
1263597046_product_3804725264.jpg'
```

Image preprocessing spec for Rendu 2:

- Standard target size: **224×224 px** (compatible with ResNet, EfficientNet, ViT pretrained weights)
- Normalization: ImageNet mean/std `[0.485, 0.456, 0.406]` / `[0.229, 0.224, 0.225]`
- Augmentation (training only): random horizontal flip, color jitter — to be defined in Rendu 2
- Image loading: PIL → torch tensor via `torchvision.transforms`

```
Saved train_cleaned.pkl : (84916, 9)
Saved test_cleaned.pkl  : (13812, 8)

Column dtypes:
productid           int64
imageid             int64
designation        object
description        object
text_clean         object
has_description     int64
word_count          int64
char_count          int64
prdtypecode         int64
dtype: object
```

5. Conclusions

Key Findings

Finding	Detail
Class imbalance	Highly skewed distribution across 27 categories (χ^2 $p \ll 0.05$). Top class is ~10× larger than the smallest. Requires weighted loss functions or class-balanced training in Rendu 2.
Missing descriptions	~35% of rows have no description — category-dependent (χ^2 independence $p \ll 0.05$), not random. Image and designation are the primary signals for those products.
Text informativeness	Word count varies significantly by category (Kruskal-Wallis $p \ll 0.05$). TF-IDF analysis reveals strong discriminative vocabulary per class.
Image diversity	Images vary in resolution and composition. Visual categories are often distinguishable but within-class variation is high. Standard 224×224 resize required for CNN models.
Multilingual text	Mostly French; some German/English. HTML noise present in descriptions — stripped during preprocessing.

Challenges for Rendu 2

1. **Class imbalance** — use `class_weight='balanced'` or weighted sampling
2. **Missing descriptions** — fallback to `designation` only; `has_description` feature flags this
3. **Multimodal fusion** — text encoder + image encoder → late fusion or joint embedding
4. **Image standardization** — all images to 224×224 px for CNN/ViT compatibility

Outlook

- **Text models**: TF-IDF + logistic regression (baseline) → CamemBERT (French BERT, advanced)

- **Image models**: ResNet50 / EfficientNet pretrained on ImageNet → fine-tuned on product images
- **Fusion**: Concatenate text + image embeddings → MLP classifier
- **Evaluation**: Weighted macro F1-score (competition metric)

Outputs saved

- `data/train_cleaned.pkl` — 84,916 rows, ready for Rendu 2 modeling
- `data/test_cleaned.pkl` — 13,812 rows, ready for inference